NEC

NONLINEAR MODEL | UPA808TC

BJT NONLINEAR MODEL PARAMETERS [1]

Parameters	Q1, Q2	Parameters	Q1, Q2
IS	8e-17	MJC	0.53
BF	128	XCJC	1
NF	1	CJS	0
VAF	17	VJS	0.75
IKF	0.18	MJS	0
ISE	3.3e-15	FC	0.37
NE	1.48	TF	8e-12
BR	9.05	XTF	11.9
NR	1.05	VTF	9.55
VAR	4.3	ITF	1.78
IKR	0.009	PTF	69.1
ISC	4e-15	TR	1e-9
NC	1.5	EG	1.11
RE	0.8	XTB	0
RB	11.1	XTI	3
RBM	2.46	KF	0
IRB	0.017	AF	1
RC	7.5		
CJE	0.415e-12		
VJE	0.68		
MJE	0.53		
CJC	0.102e-12		
VJC	0.29		

(1) Gummel-Poon Model

UNITS

Parameter	Units
time	seconds
capacitance	farads
inductance	henries
resistance	ohms
voltage	volts
current	amps

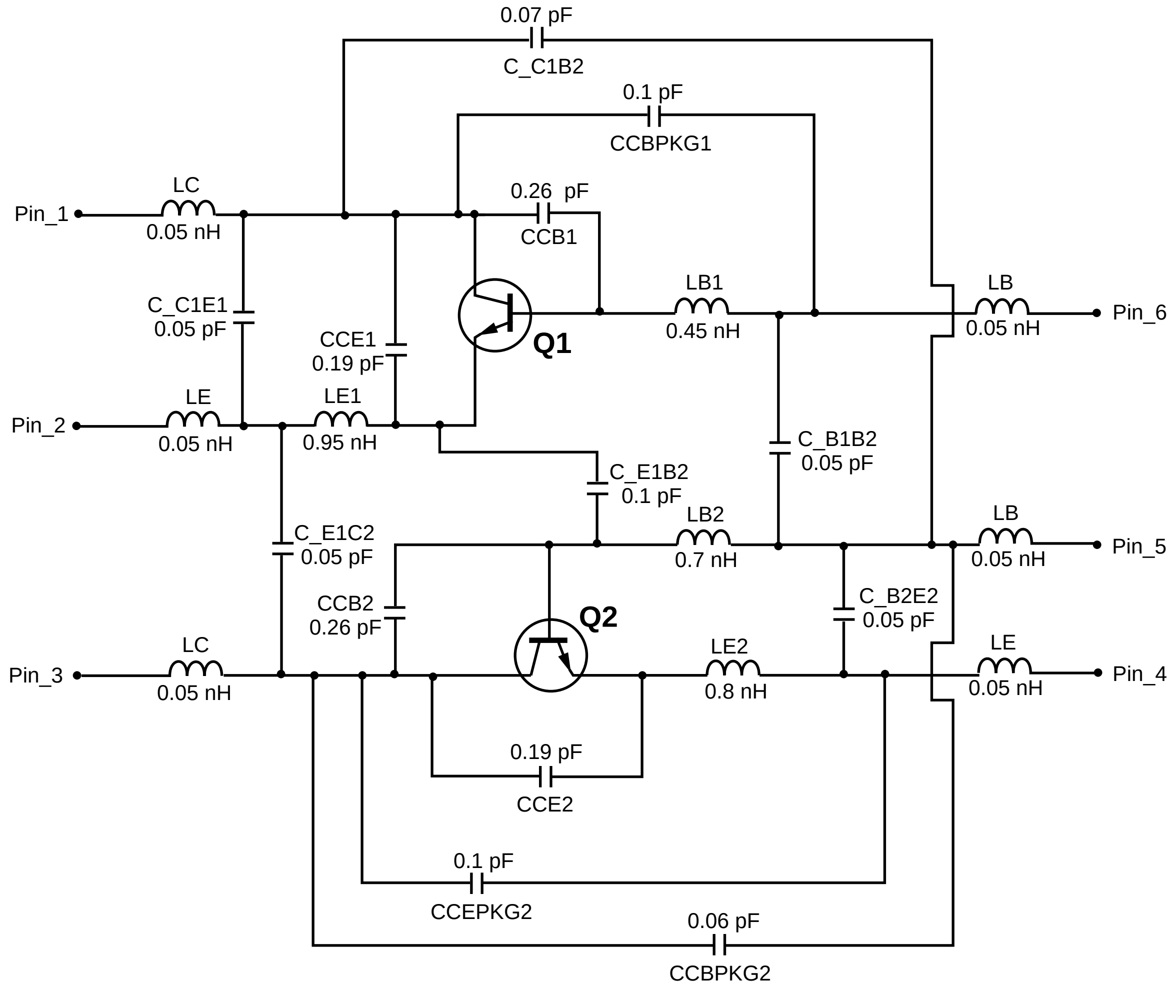
MODEL RANGE

Frequency: 0.1 to 7.0 GHz
Bias: V_{CE} = 0.5 V to 2 V, I_C = 0.5 mA to 10 mA
Date: 02/01

Note:
This nonlinear model utilized the latest data available.
See our Design Parameter Library at **www.cel.com** for this data.

NONLINEAR MODEL

SCHEMATIC



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